

# RAVEENDRA SIRAPARAPU

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## Summary

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A computer science undergraduate with hands-on experience building **API-driven systems**, **Full stack systems**, and **automation workflows**. Proficient in **Python** and the **MERN stack**. Looking for a software role to contribute meaningfully for the team.

## Education

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| <b>Sasi Institute of Technology &amp; Engineering</b>        | 2022 – 2026 |
| B.Tech in Computer Science & Engineering (CGPA: 7.35 / 10.0) |             |
| <b>Govt Junior College, Pentapadu</b>                        | 2020 – 2022 |
| Intermediate Education                                       |             |

## Experience

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| <b>National Institute of Technology, Andhra Pradesh (NIT-AP)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>May 2025 – Aug 2025</b> |
| <i>AI Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Tadepalligudem, AP</i>  |
| <ul style="list-style-type: none"><li>Contributed to the development of an <b>AI-powered chatbot</b> aimed at assisting students by improving accessibility and learning support.</li><li>Designed and implemented a <b>clean and user-friendly UI</b>, enhancing usability and overall user experience of the chatbot interface.</li><li>Integrated a <b>code compiler and execution environment</b> within the chatbot, enabling real-time code execution and increasing system interactivity.</li></ul> |                            |

## Projects

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**AI-Based Two-Wheeler Traffic Violation Detection for Road Safety** — Python, YOLOv8, YOLOv11, Flask

- Developed an AI-based traffic violation detection system capable of identifying helmet absence, triple riding, mobile phone usage, wheeling, and no parking violations using computer vision techniques.
- Implemented and compared YOLOv8 and YOLOv11 object detection models to analyze performance based on accuracy, precision, recall, F1-score, and confusion matrix evaluation metrics.
- Integrated OpenCV-based image preprocessing and real-time bounding box visualization to improve detection accuracy and enhance violation identification.
- Designed a web-based interface using Flask, HTML, CSS, and JavaScript to enable users to upload traffic images and receive automated violation detection results in real time.

**IPL Match Prediction Web Application** — React, HTML, CSS, Python, REST APIs, Machine Learning

- Developed a full-stack IPL match prediction web application using React for the frontend and Python for backend processing.
- Designed and integrated RESTful APIs to handle match data input, prediction requests, and result delivery in real time.
- Implemented machine learning based prediction logic using historical IPL datasets to estimate match-winning probabilities.
- Built an interactive and responsive user interface for entering match parameters and visualizing prediction results effectively.

## Technical Skills

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**Programming Languages:** Python  
**Frontend:** HTML, CSS, JS, React.js  
**Backend:** Node.js, Express.js, RESTful APIs  
**Databases:** MongoDB, SQL  
**Libraries:** NumPy, Pandas, Matplotlib  
**DevOps & Tools:** Git, GitHub, Docker, Jenkins, AWS (S3 & EC2) Postman, VS Code  
**Coursework:** Data Structures, DBMS, Data Mining

## Achievements

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- NPTEL Certified – The Joy of Computing in Python (68%):** Demonstrated strong foundations in problem-solving and Python-based algorithmic thinking.
- NPTEL Certified – Python for Data Science (64%):** Gained hands-on experience in data analysis, visualization, and numerical computing workflows.
- HackerRank Certified – Python (Basic) & Java (Basic):** Validated core programming concepts, control structures, and object-oriented fundamentals through standardized assessments.